UNDERTAKING

I, hereby declare that I have completed the research work for the full time period prescribed under the clause VIII.1 of the Ph.D. ordinances of the Banaras Hindu University, Varanasi and that the research work embodied in this thesis entitled **"Supersymmetry and Shape Invariance of Exceptional Orthogonal Polynomials and Rational Extension of Various Quantum Mechanical Systems"** is my own research work.

I avail myself to responsibility such as an act will be taken on behalf of me, mistakes, errors of fact and misinterpretation are of course entirely my own.

Date: 22-08-2024

Place: Varanasi

Satish Yadav

(Satish Yadav)

ANNEXURE - G
[Clause XIII.2 (b) (iii)]

CANDIDATE'S DECLARATION

I, ***Satish Yadav***, s/o Mr. Ramashankar Yadav, certify that the work embodied in this Ph.D. thesis is my own bonafide work carried out by me under the supervision of ***Prof. Bhabani Prasad Mandal***, Department of Physics, B.H.U. for a period of ***March 2019*** to ***August 2024*** at Banaras Hindu University, Varanasi. The matter embodied in this Ph.D. thesis has not been submitted for the award of any other degree/diploma.

I declare that I have devotedly acknowledged, given credit and refereed to the research workers wherever their works have been cited in the text and the body of the thesis. I further certify that I have not willfully lifted up some other's work, para, text, data, results, etc. reported in the journals, books, magazines, reports, dissertations, theses, etc., or available at web-sites and included them in this Ph.D. thesis and cited as my own work.

Date: 22-08-2024
Place: Varanasi

Satish Yadav
(Satish Yadav)

Certificate from the Supervisor/Co-supervisor

This is to certify that the above statement made by the candidate is correct to the best of my knowledge.

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ANNEXURE - E
[Clause XIII.1 (c) & XIII.2 (b) (iv)]

COURSE WORK COMPLETION CERTIFICATE

This is to certify that **Satish Yadav**, a bonafide research scholar of this department has successfully completed the course work requirement, which is a part of his Ph.D. programme.

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ANNEXURE - E
[Clause XIII.1 (c) & XIII.2 (b) (iv)]

PRE-SUBMISSION SEMINAR CERTIFICATE

This is to certify that **Satish Yadav**, a bonafide research scholar of this department, has successfully completed the pre-submission seminar requirement on topic "**Supersymmetry and Shape Invariance of Exceptional Orthogonal Polynomials and Rational Extension of Various Quantum Mechanical Systems**" dated **August 8th, 2024**, which is a part of his Ph.D. programme.

R. Singh

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Date: 22-08-2024

Place: Varanasi

ANNEXURE - H
[Clause XIII.2 (b) (v)]

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Candidate's Name: Satish Yadav

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Satish Yadav
(Satish Yadav)

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Dedicated to my family & friends

List of Publications

Referred Journals
(Relevant to the present Thesis)

1. **Supersymmetry and Shape Invariance of Exceptional Orthogonal Polynomials**
   Satish Yadav, Avinash Khare and B. P. Mandal
   Annals of Physics 444, 169064 (2022)
   DOI: https://doi.org/10.1016/j.aop.2022.169064

2. **New solutions of Isochronous potentials in terms of exceptional orthogonal polynomials in heterostructures**
   Satish Yadav, Rahul Ghosh and B. P. Mandal
   arXiv preprint arXiv:2401.00995(2024)
   DOI: https://doi.org/10.48550/arXiv.2401.00995

3. **Generating QES potentials supporting zero energy normalizable states for an extended class of truncated Calogero Sutherland model**
   Satish Yadav, Sudhanshu Shekhar, Bijan Bagchi and B. P. Mandal
   arXiv preprint arXiv:2406.09164(2024)
   DOI: https://doi.org/10.48550/arXiv.2406.09164

4. **Supersymmetry and shape invariance of exceptional Hermite polynomials**
   Satish Yadav, Avinash Khare and B. P. Mandal
   (To be submitted)

5. **Master function associated with exceptional orthogonal polynomials**
   Satish Yadav, Avinash Khare and B. P. Mandal
   (In prepration)

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(Satish Yadav)

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